

Regulatory-Ready Platform

for Multiomics Diagnostics

COMMERCIAL BRIEFING

From discovery scripts to auditor-ready evidence — without rebuilding your stack for every assay.

The problem buyers already feel

Failure mode	What happens today
Slow path to market	Biomarker → 510(k) / De Novo / PMA (+ CLIA) stalls on process, not only science
Quiet validation failure	Train/holdout leakage, notebooks, unreproducible “final” runs
Audit bottleneck	Documentation debt — not missing compute
Modality sprawl	Each omics or indication becomes a new product stack

MethylPipeline shrinks those modes with config-first science, partition discipline, and submission scaffolds.

Claim boundary

The platform **enforces process and configuration controls** and **assembles evidence packages**.
Architecture and SaMD profiles are **not** FDA clearance or approval.

We do	We do not
Shorten the software + validation process path	Choose or guarantee the regulatory pathway
Lock partitions, configs, and provenance	Replace clinical judgment or predicate strategy
Stage clinical-performance claim gates	Imply every assay is a simple 510(k)

Positioning

Control plane + process packs + application packs

Bridges **bioinformatics R&D engineering** and **regulatory / quality readiness**.

Three pillars of value

1. R&D agility

Cohorts, ECDF / tabular models, covariates, Houseman / HiTIMED — via **configuration**, not ad hoc scripts.

2. Cloud efficiency

Capability-matched workers + **CAAS** skip alignment when only downstream science changes.

3. SaMD controls

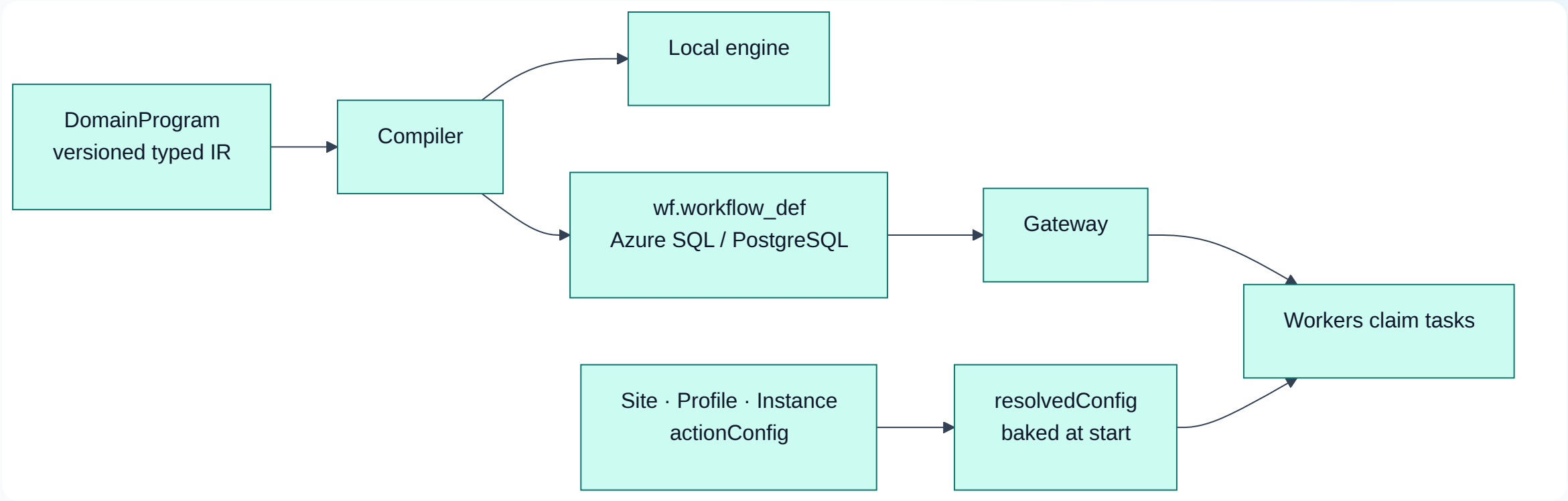
Partition discipline, staged clinical-performance claims, and auto-assembled **evidence scaffolds**.

What buyers actually purchase

Layer	What it is
1. Execution platform	DomainProgram IR, typed actions, cfg/wf registry, portal, gateway, CAAS, fleet control
2. Process packs	Modality science — methylation WGBS, RNA-Seq, proteomics
3. Application packs	Indication / trait overlays as config on an existing process

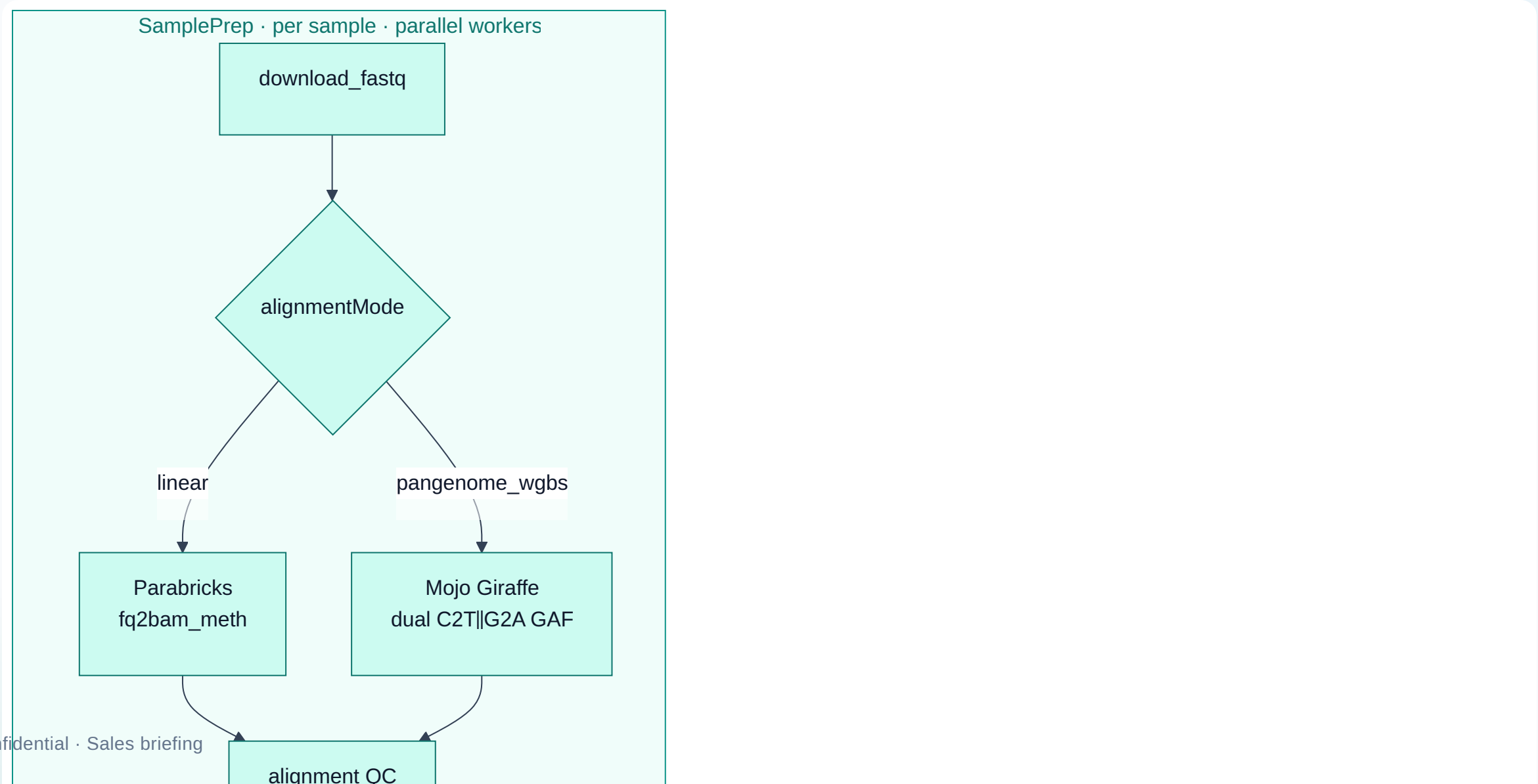
- DISEASE-AGNOSTIC
- ANALYTE-EXTENSIBLE
- OPERATOR-CONFIGURABLE
- CONFIG NOT CODE

DomainProgram: declare the science once



Strength	Detail
One language	SamplePrep, MC stability, freeze, model selection, holdouts
Config layers	Operators tune parameters without editing Python
Worker contract	Tasks carry baked <code>resolvedConfig</code> — no profile re-read on the node

Methylation process pack (production path)

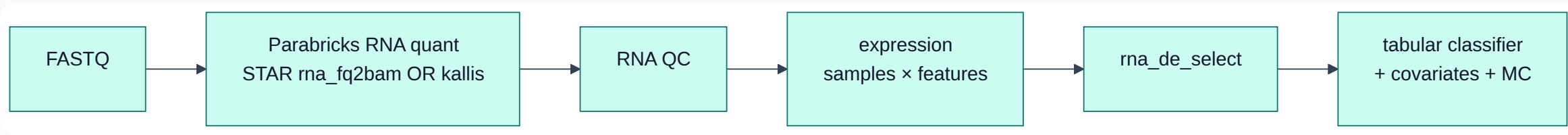


GPU pangenome WGBS (current engineering)

Capability	Production contract
Dual-graph Align	C2T G2A → science GAF for MethylCall
Engine knobs	<code>actionConfig.methylgrapher_wgbs</code> → task <code>resolvedConfig</code>
Backends	<code>gpu_giraffe</code> / <code>mojo_giraffe</code> / <code>cpu_vg</code> (operator-set)
Image	<code>epimethyl/methylgrapher:1.70-mojo</code> on GH200-class fleet
QC path	Mojo QC BAM + conversion-rate sidecars when enabled

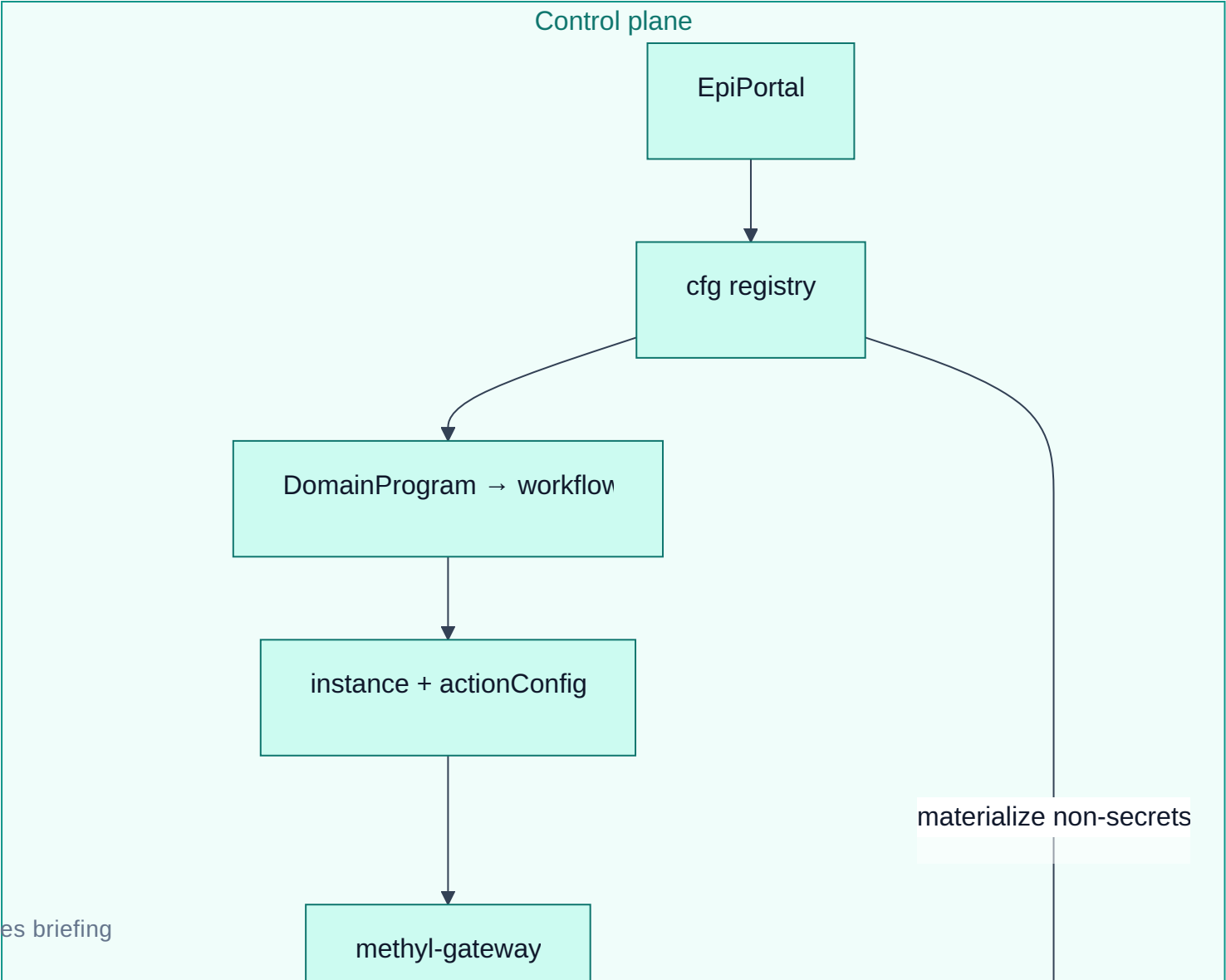
Linear `fq2bam_meth` remains the comparator; **pangenome_wgbs** is the graph science path — not a separate product.

Same language, RNA-Seq process pack



Shared	Pack-specific
Scheduler, cfg/wf, CAAS, feature seam	<code>actionConfig.rna_align.quant_mode</code>
Portal / gateway / workers	RNA QC thresholds
Tabular model MC	STAR vs kallisto branch

Distributed execution



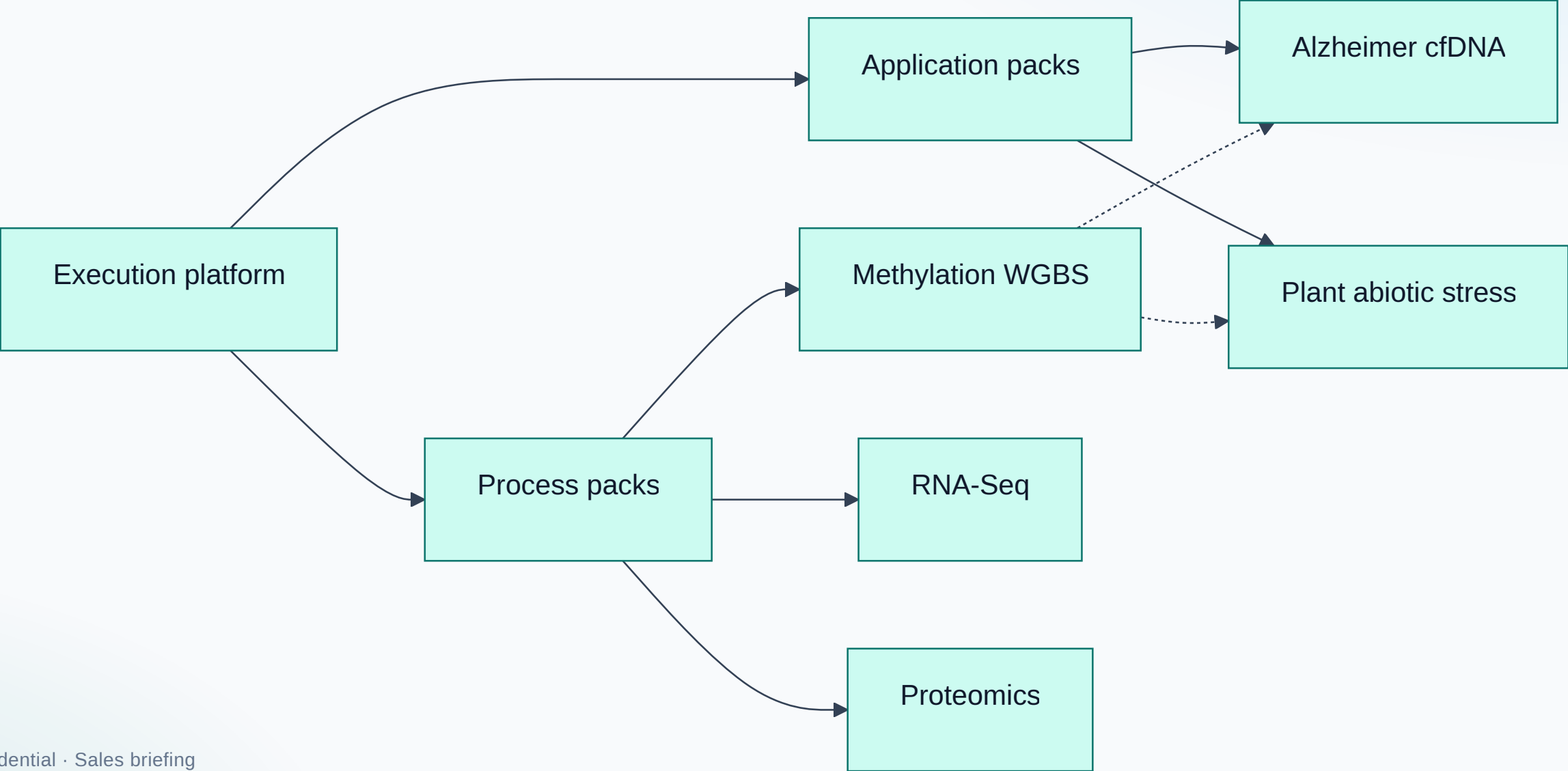
Config layers — tune without redeploying science code



Layer	Who edits	Examples
Site	Operators / cluster	genomes, caches, Align image, fleet caps
Profile	Release / procedure	<code>samd_research</code> , MC knobs, gene FeatureCuts
DomainProgram	Pack authors	topology, typed actions
Study / instance	Study team	cohorts, partitions, claim gates

Merge: site → profile → program/instance overlay → analyte fill-missing → *(no Python fallback for tunable knobs)*.

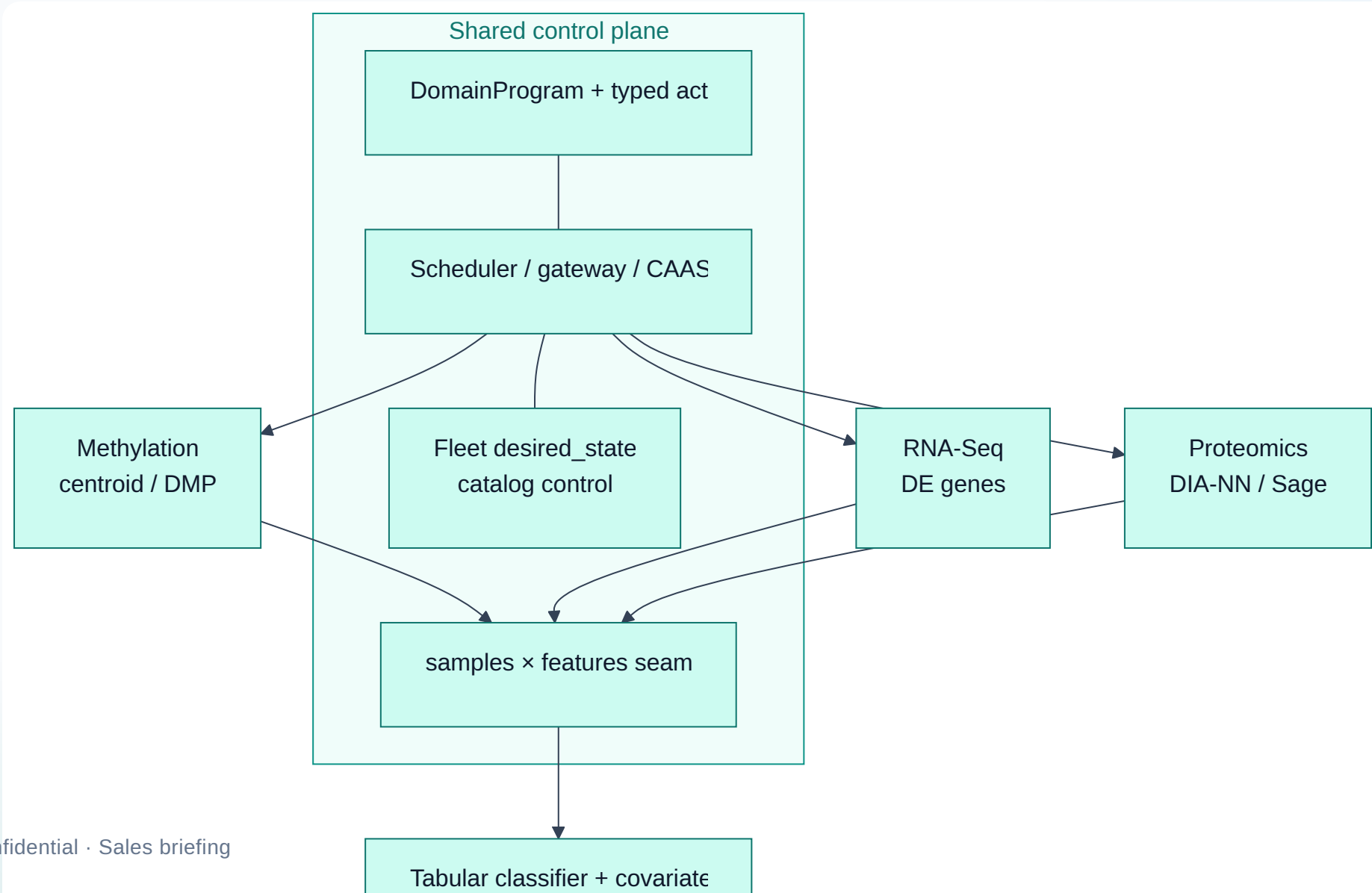
Process packs vs application packs



Analyte & disease-process roadmap

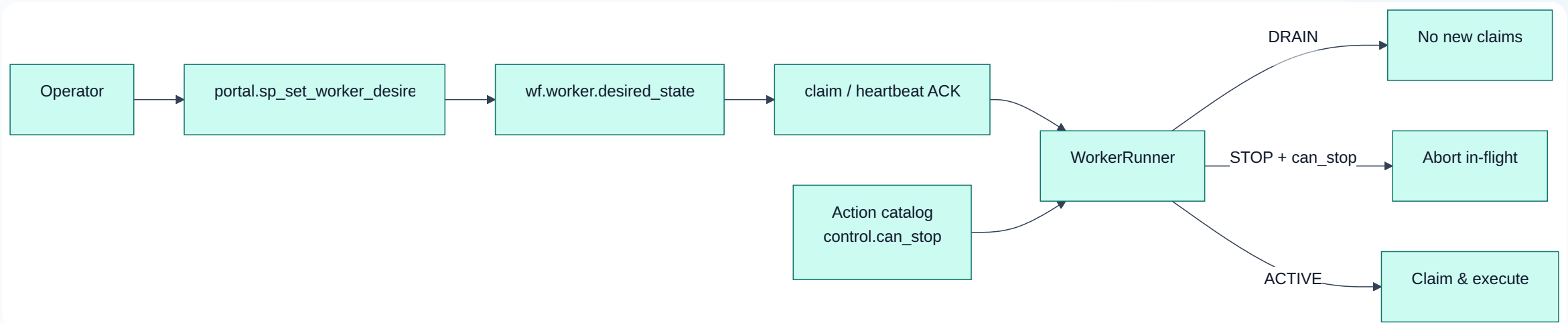
Process / analyte	Status	What ships
Methylation — buffy / cfDNA	Production	SamplePrep → MC → freeze → model; SaMD ladder; Mojo pangenome_wgbs
RNA-Seq	Research pack	STAR/kallisto, RNA QC, DE panel → tabular classifier
Proteomics	Research pack	GPU DIA-NN, Sage DDA, panel ingest; Prosit / Casanovo
Alzheimer cfDNA	Research app pack	Methylation + neuro-core overlay — config only
Plant abiotic stress	Research app pack	Control vs Drought WGBS; multi-crop sites

Multiomics on one plane



Fleet control without SSH

Operators set worker state from the portal / SQL — workers honor it on claim and heartbeat.



desired_state	Idle	In-flight
ACTIVE	Claim	Continue
DRAINING	Skip claims	Finish current (pause only if can_pause)
STOPPING	Skip claims	Abort if catalog can_stop

Science knobs stay on `actionConfig`. Fleet control stays on worker state + catalog `control`.

Application pack pattern

Alzheimer cfDNA

`primary_analyte: cfdna`

Patient-disjoint partitions; Alzheimer's disease term; `neuro-core`; Control → MCI → AD.

No new actions or aligners.

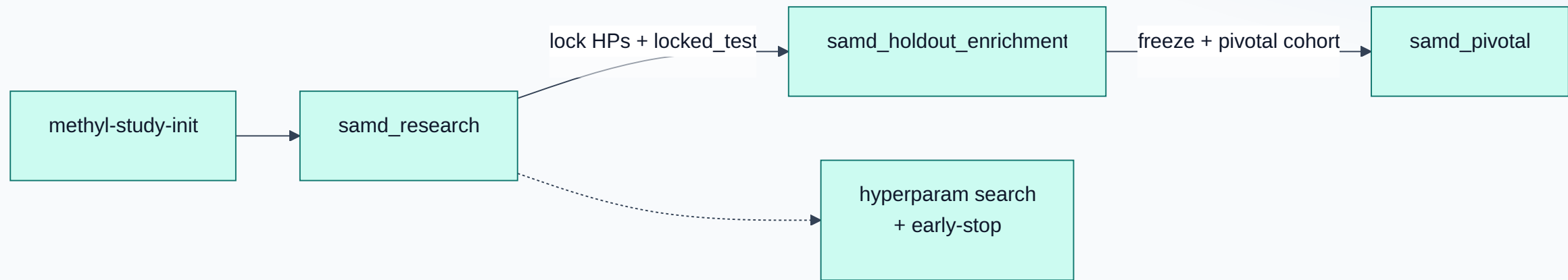
Plant abiotic stress

`primary_analyte: plant_tissue`

Control vs Drought across CG/CHG/CHH; Ensembl Plants sites; offline trait priors.

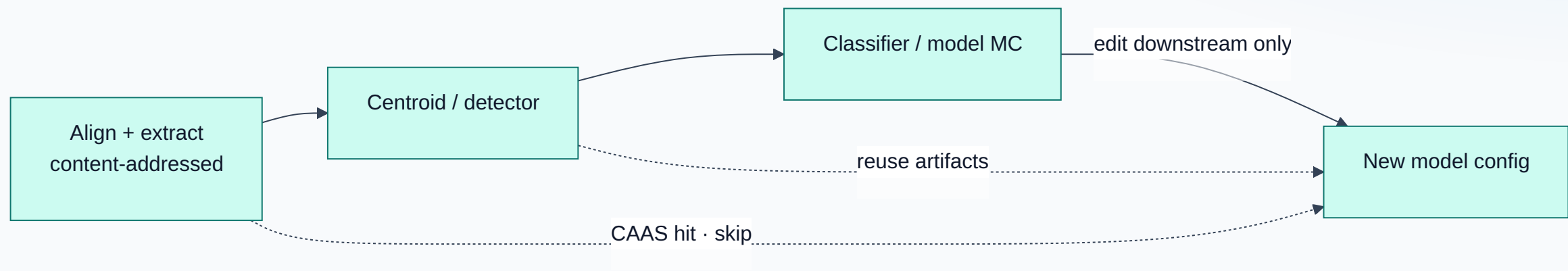
epi-GBS isolated via separate SamplePrep program.

SaMD profile ladder



Profile	Intent
samd_research	Discover stable panels under partition discipline
samd_holdout_enrichment	Locked hyperparameters + locked test patients
samd_pivotal	Claim-gated clinical-performance evidence packaging

CAAS: change the classifier, keep the alignment

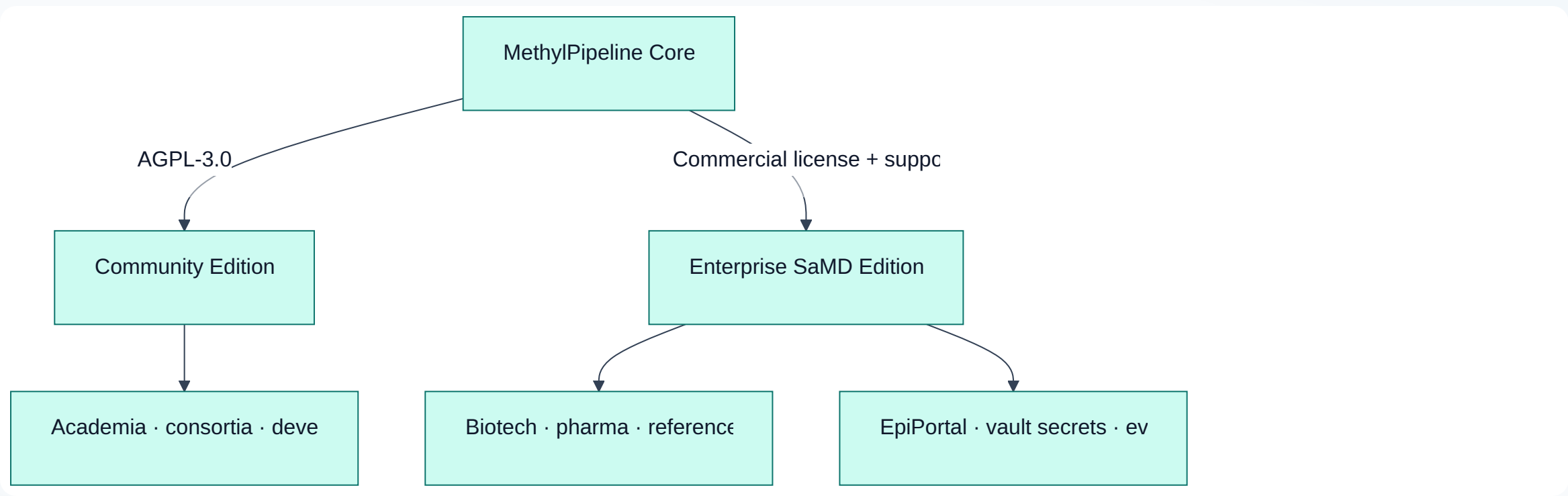


Mechanism	Why it matters
Input / output signatures	Skip recomputation when science is unchanged
CLI + environment binding	Provenance for quality review
FOREACH bundle keys	Whole MC iteration short-circuit when safe

Packaging & GTM

Open core that captures R&D — Enterprise that closes the FDA chasm

Open-core packaging



Community vs Enterprise

Community Edition

AGPL-3.0

- LocalWorkflowEngine
- DNA methylation process pack
- CLI tools + `samd_research`
- Goal: citations, adoption, contributions

Enterprise / SaMD Edition

ANNUAL SUBSCRIPTION

- Evidence index + traceability packaging
- Pivotal ladder unlocks
- EpiPortal multi-tenant ops + fleet control
- Zero-trust credentials + GPU SamplePrep support

Hybrid cloud (BYOC)

We host / operate

- EpiPortal control plane
- DB metadata + scheduler
- Non-secret materialization

Customer keeps

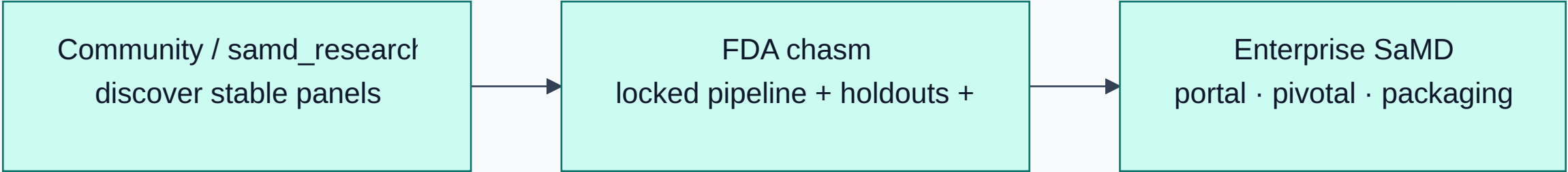
- Stateless workers (AWS / Azure / on-prem)
- `/work` + samples in their boundary
- HIPAA / GDPR data residency

Regulated data never has to leave the customer's estate.

Who buys — and why

Buyer	Why MethylPipeline
Biotech / diagnostic startups	Liquid biopsy & multiomics without building a platform
Pharma / trial sponsors	Locked, reproducible workflows as trial endpoints
Reference labs / CROs	FASTQs → curated classification + evidence packages

Customer journey: the “FDA chasm”



Stage	Motion
Hook	Research feasibility at low friction
Trigger	Marker must survive locked + pivotal validation under audit
Upgrade	Portal ops, pivotal ladder, regulatory packaging

Why technical buyers say yes

Capability	Outcome
Audit trails with low overhead	Hashes, versions, environment signatures in action results
CAAS economics	Skip alignment across large cohorts when iterating models
Multionics extensibility	Process packs add programs; application packs are config-only
Partition discipline	Development vs locked holdout enforced by profiles
Fleet control	Drain / stop workers from portal — no SSH to GPU nodes
Config not code	Science knobs in site/profile/instance — never magic numbers in Python

Close

Regulatory-Ready Platform for Multiomics Diagnostics

Control plane + packs — not a single hard-coded assay.

Next step: live DomainProgram demo · SaMD ladder walkthrough · evidence-scaffold review

